Instance Details:

Instance summary for i-062c578fcb061ba43 (joao-safe-server) Info

Connect | Instance state | Actions

Updated 1 minute ago

Instance ID i-062c578fcb061ba43	**Public IPv4 address** 54.167.24.198 | open address	**Private IPv4 addresses** 172.31.17.84
IPv6 address –	**Instance state** Running	**Public DNS** ec2-54-167-24-198.compute-1.amazonaws.com | open address
Hostname type IP name: ip-172-31-17-84.ec2.internal	**Private IP DNS name (IPv4 only)** ip-172-31-17-84.ec2.internal	
Answer private resource DNS name IPv4 (A)	**Instance type** t2.micro	**Elastic IP addresses** –
Auto-assigned IP address 54.167.24.198 [Public IP]	**VPC ID** vpc-0c191489f702038c7 (default)	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-04e5e8b3ad0f851a7	**Auto Scaling Group name** –
IMDSv2 Required	**Instance ARN** arn:aws:ec2:us-east-1:686699774218:instance/i-062c578fcb061ba43	**Managed** false
Operator –		

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Security details

IAM role –	**Owner ID** 686699774218	**Launch time** Fri Apr 10 2026 13:11:01 GMT+0100 (Hora de verão da Europa Ocidental)
Security groups sg-0ebaf0a77e3c5260f (dns-joao)		

Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
–	sgr-05b2f3ab11ff48d30	80	TCP	0.0.0.0/0	dns-joao	–
–	sgr-0580d60f61ea898a3	22	TCP	0.0.0.0/0	dns-joao	–
–	sgr-06e773a9a21a4b33f	443	TCP	0.0.0.0/0	dns-joao	–

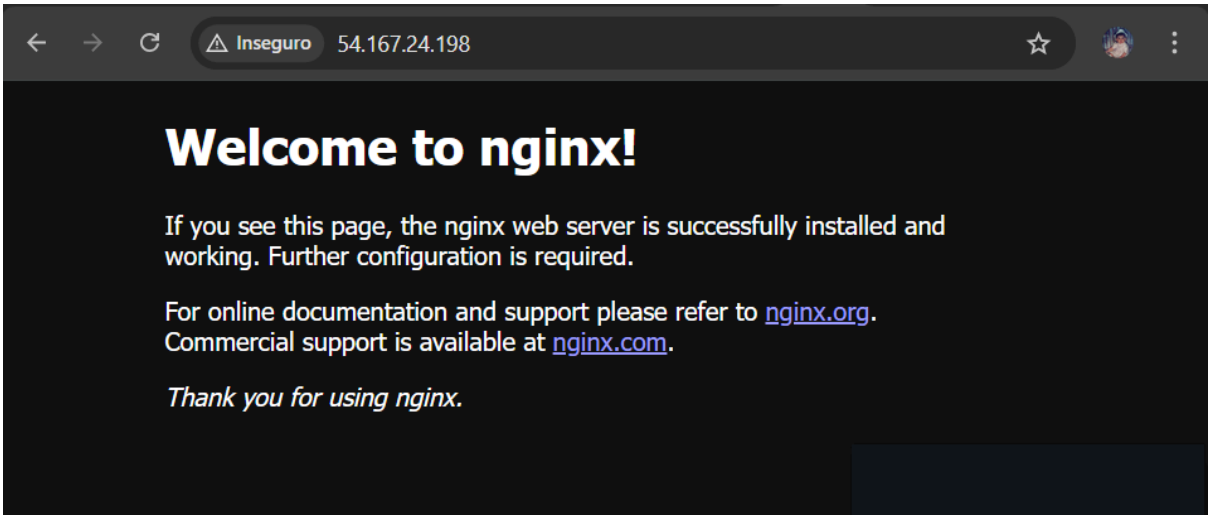
Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
–	sgr-07c3aeacc9f372f12	All	All	0.0.0.0/0	dns-joao	–

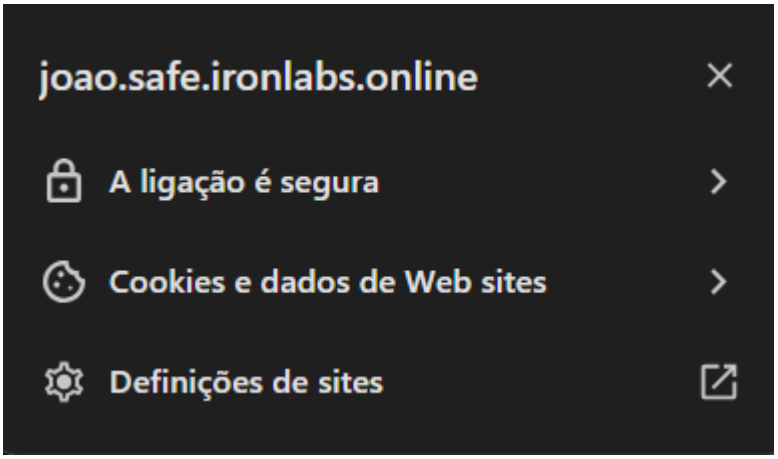
EC2 Public IP on Nginx browser:

```
ubuntu@ip-172-31-17-84:~$ curl http://54.167.24.198
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installe
d and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

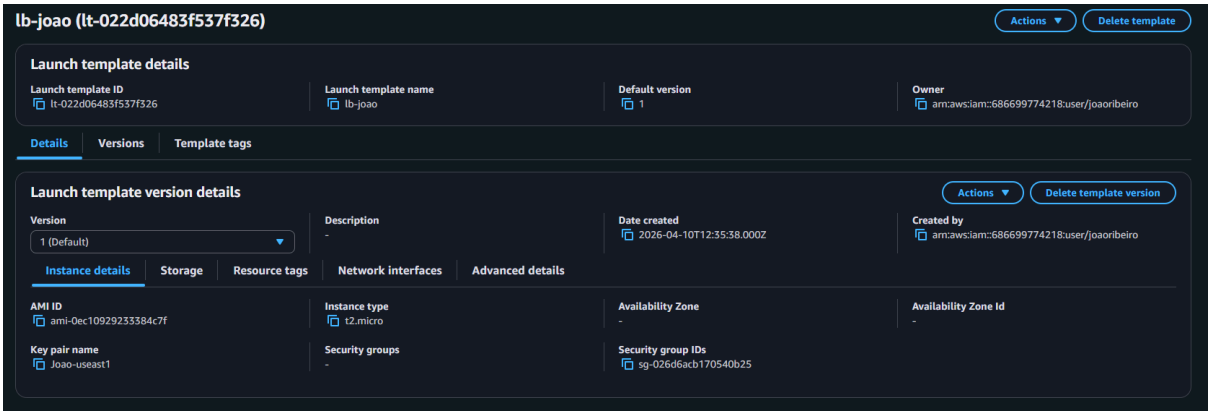


Certbot Success Certification:

```
ubuntu@ip-172-31-17-84:~$ sudo certbot --nginx -d joao.safe.ironlabs.o
nline
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for joao.safe.ironlabs.online

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/joao.safe.ironlabs.onli
ne/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/joao.safe.ironlabs.onli
ne/privkey.pem
This certificate expires on 2026-07-09.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certif
icate in the background.

Deploying certificate
Successfully deployed certificate for joao.safe.ironlabs.online to /et
c/nginx/sites-enabled/default
Congratulations! You have successfully enabled HTTPS on https://joao.s
afe.ironlabs.online

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
- - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
- - - - -
```



I created an launch template that help me to create the two instances for the project:



The two instances running:

joao-LB-Server-2	i-0cc5778a2014c072d	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-98-80-74-52.comp...	98.80.74.52	–
joao-LB-Server-1	i-0984aeec9c6832846	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-100-31-108-189.co...	100.31.108.189	–

Instance summary for i-0984aeec9c6832846 (joao-LB-Server-1) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID i-0984aeec9c6832846	**Public IPv4 address** 100.31.108.189 | open address	**Private IPv4 addresses** 172.31.29.208
IPv6 address –	**Instance state** Running	**Public DNS** ec2-100-31-108-189.compute-1.amazonaws.com | open address
Hostname type IP name: ip-172-31-29-208.ec2.internal	**Private IP DNS name (IPv4 only)** ip-172-31-29-208.ec2.internal	
Answer private resource DNS name –	**Instance type** t2.micro	**Elastic IP addresses** –
Auto-assigned IP address 100.31.108.189 [Public IP]	**VPC ID** vpc-0c191489f702038c7 (default)	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-04e5e8b3ad0f851a7	**Auto Scaling Group name** –
IMDSv2 Required	**Instance ARN** arn:aws:ec2:us-east-1:686699774218:instance/i-0984aeec9c6832846	**Managed** false
Operator –		

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance details Info

AMI ID ami-0ec10929233384c7f	**Monitoring** disabled	**Platform details** Linux/UNIX
AMI name ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20260313	**Allowed image** -	**Termination protection** Disabled

Instance summary for i-0cc5778a2014c072d (joao-LB-Server-2) Info

Connect | Instance state | Actions

Updated 2 minutes ago

Instance ID i-0cc5778a2014c072d	**Public IPv4 address** 98.80.74.52 | open address	**Private IPv4 addresses** 172.31.17.21
IPv6 address –	**Instance state** Running	**Public DNS** ec2-98-80-74-52.compute-1.amazonaws.com | open address
Hostname type IP name: ip-172-31-17-21.ec2.internal	**Private IP DNS name (IPv4 only)** ip-172-31-17-21.ec2.internal	
Answer private resource DNS name –	**Instance type** t2.micro	**Elastic IP addresses** –
Auto-assigned IP address 98.80.74.52 [Public IP]	**VPC ID** vpc-0c191489f702038c7 (default)	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-04e5e8b3ad0f851a7	**Auto Scaling Group name** –
IMDSv2 Required	**Instance ARN** arn:aws:ec2:us-east-1:686699774218:instance/i-0cc5778a2014c072d	**Managed** false
Operator –		

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance details Info

AMI ID ami-0ec10929233384c7f	**Monitoring** disabled	**Platform details** Linux/UNIX

Certificate Accepted:

e4c60eff-583b-4293-b0d4-7b5b36b7acfe Delete

Certificate status

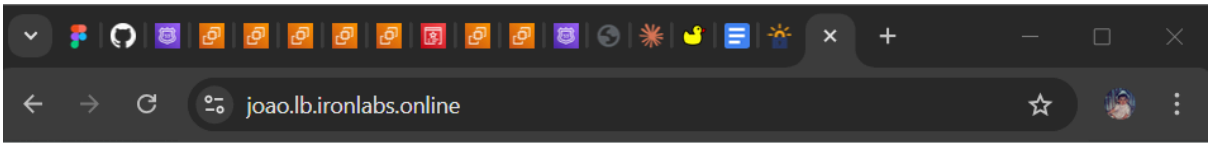
Identifier
e4c60eff-583b-4293-b0d4-7b5b36b7acfe

Status
Issued

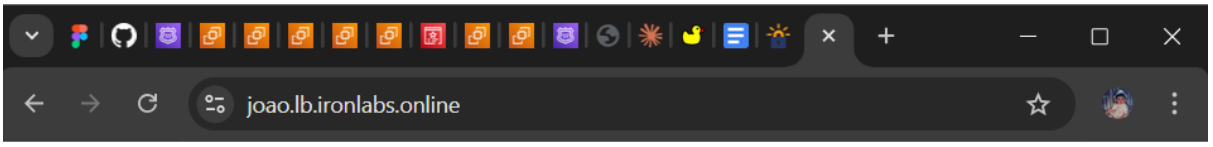
ARN
arn:aws:acm:us-east-1:686699774218:certificate/e4c60eff-583b-4293-b0d4-7b5b36b7acfe

Type
Amazon Issued

https://joao.lb.ironlabs.online



Hello from joao-LB-Server-1



Hello from joao-LB-Server-2

–